

Electronic Devices

Final Term Lecture - 01

Reference book:

**Electronic Devices and Circuit Theory
(Chapter-5)**

Robert L. Boylestad and L. Nashelsky , (11th Edition)



OBJECTIVES

- Become familiar with the re , hybrid, and hybrid p models for the BJT transistor.
- Learn to use the equivalent model to find the important ac parameters for an amplifier.
- Understand the effects of a source resistance and load resistor on the overall gain and characteristics of an amplifier.
- Become aware of the general ac characteristics of a variety of important BJT configurations.
- Begin to understand the advantages associated with the two-port systems approach to single- and multistage amplifiers.
- Develop some skill in troubleshooting ac amplifier networks.

BJT TRANSISTOR MODELING

- A **model** is an equivalent circuit that represents the **AC characteristics** of the transistor.
- A **model** uses circuit elements that **approximate the behavior** of the transistor.
- There **are two models commonly used** in small signal AC analysis of a transistor:
 - r_e model
 - Hybrid equivalent model

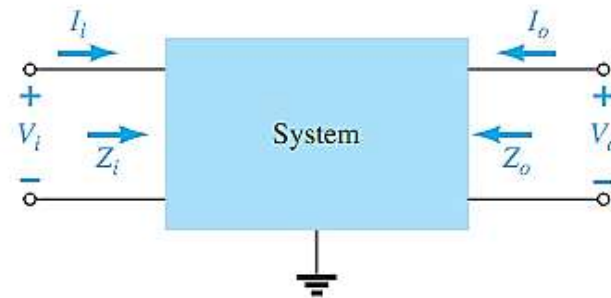
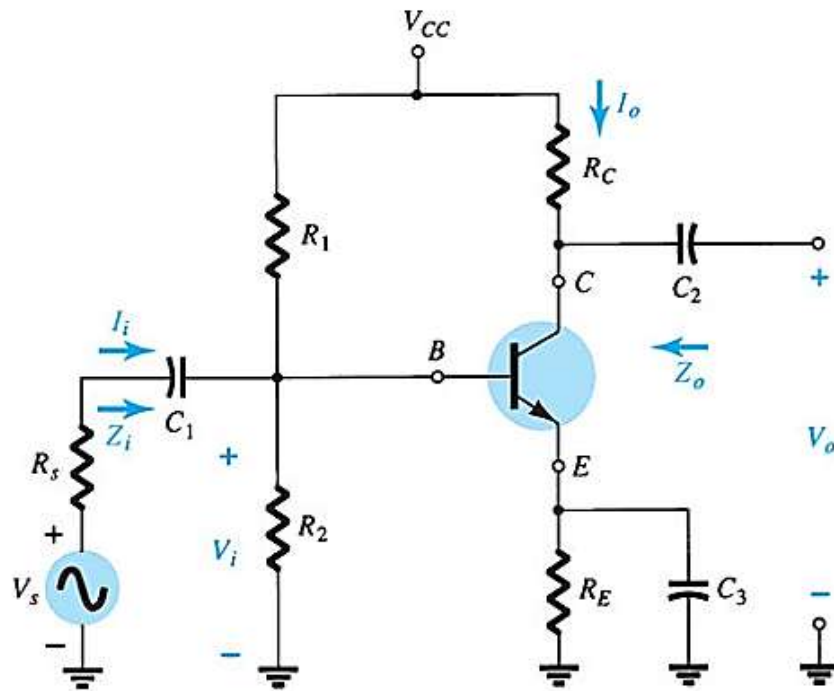


FIG. 5.5

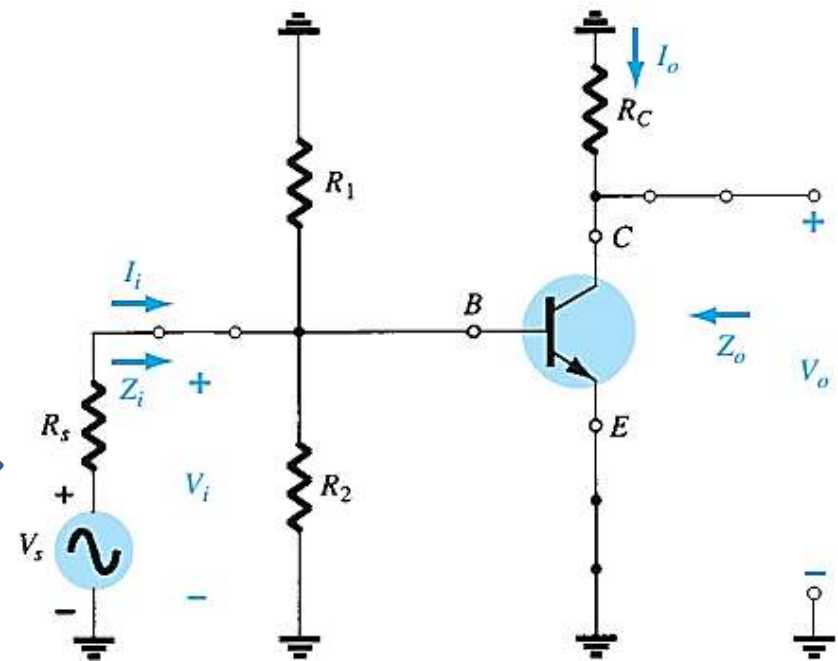
Defining the important parameters of any system.

BJT TRANSISTOR MODELING

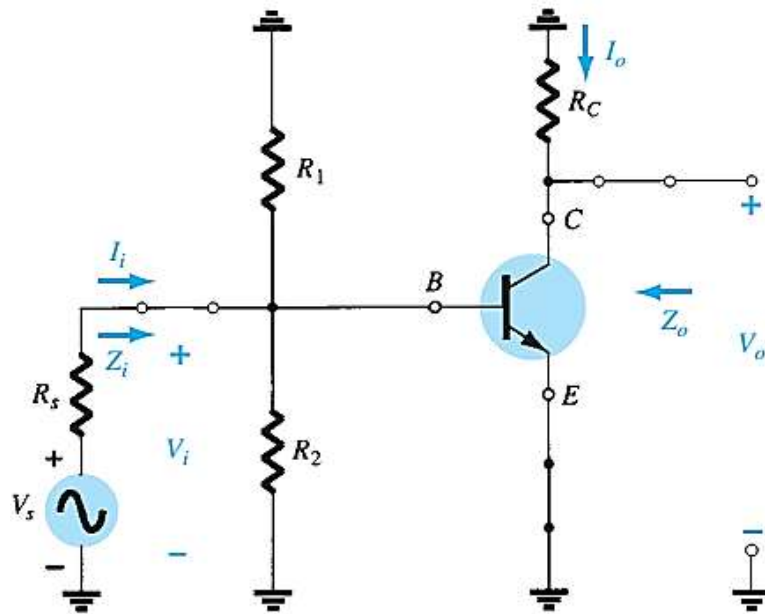


Capacitors chosen with **very small reactance** at the frequency of application → **replaced by low-resistance or short circuit.**

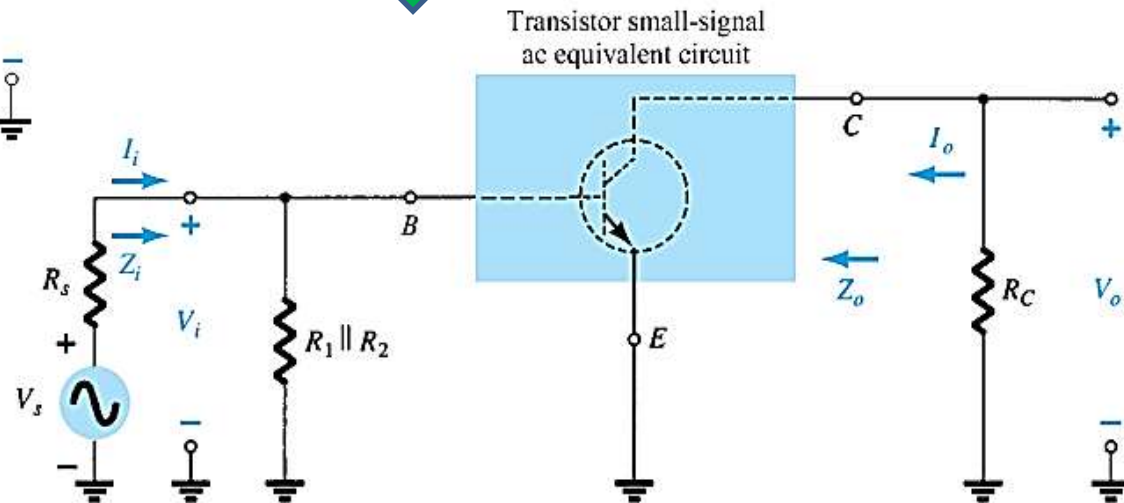
Removal of the dc supply and insertion of the short-circuit equivalent for the capacitors.



BJT TRANSISTOR MODELING

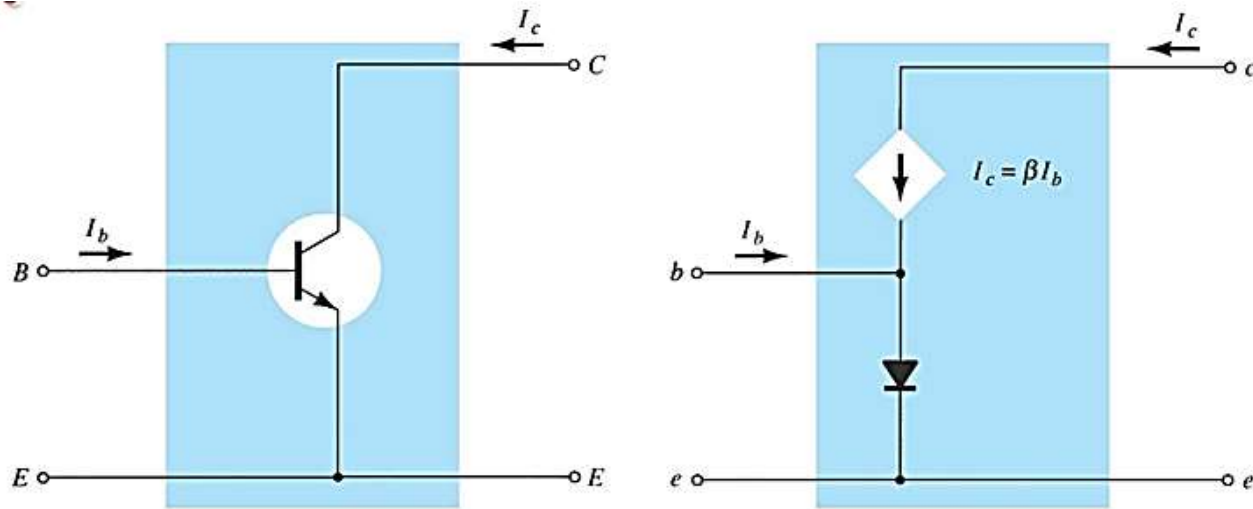


Circuit redrawn for small
signal ac analysis



Transistor small-signal
ac equivalent circuit

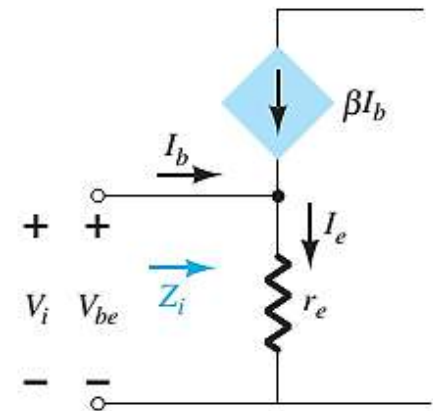
The r_e Transistor Model (Common Emitter Configuration)



$$Z_i = \frac{V_i}{I_b} = \frac{V_{be}}{I_b}$$

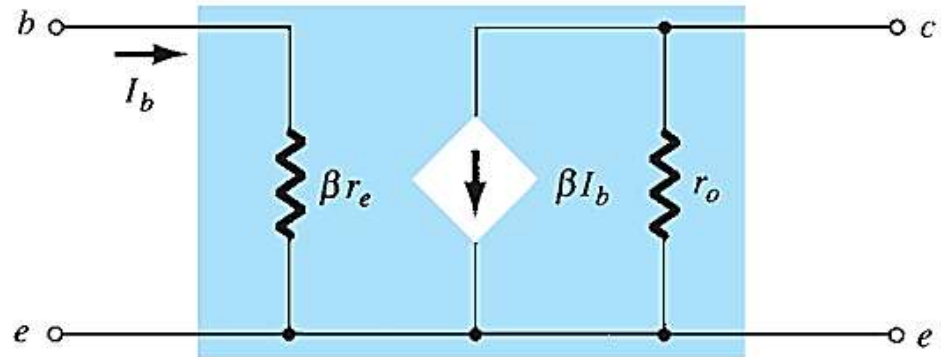
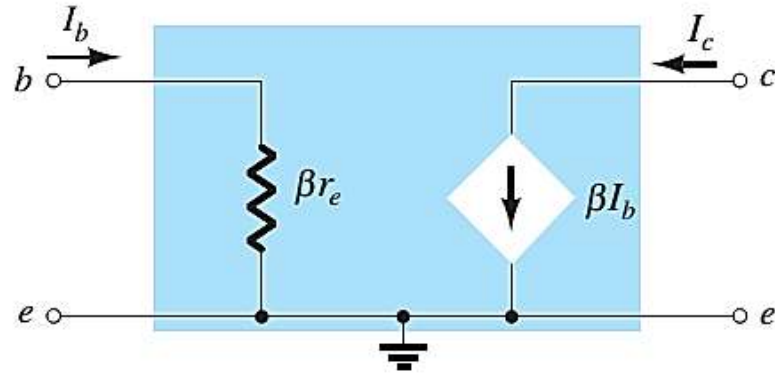
$$V_{be} = I_e r_e = (I_c + I_b) r_e = (\beta I_b + I_b) r_e \\ = (\beta + 1) I_b r_e$$

$$Z_i = \frac{V_{be}}{I_b} = \frac{(\beta + 1) I_b r_e}{I_b} = (\beta + 1) r_e \approx \beta r_e$$



The r_e Transistor Model (Common Emitter Configuration)

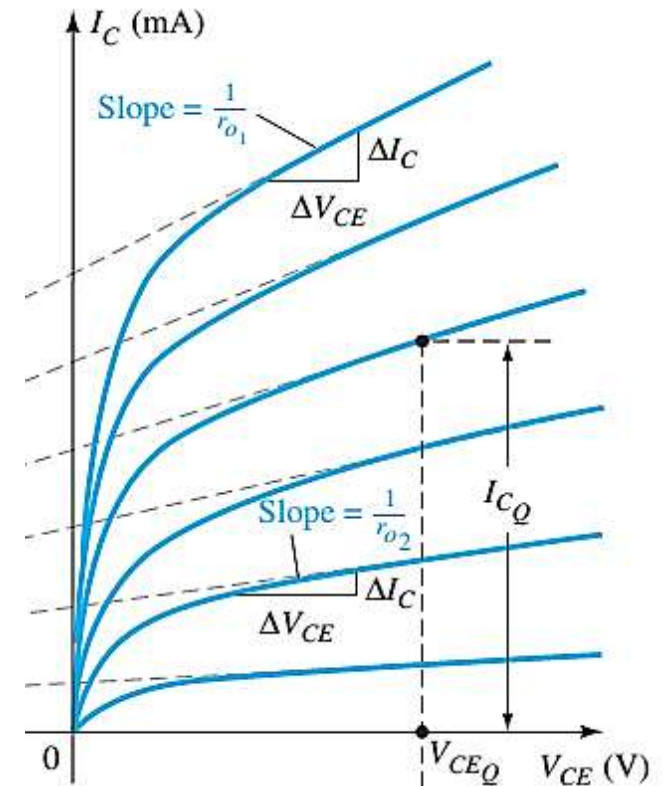
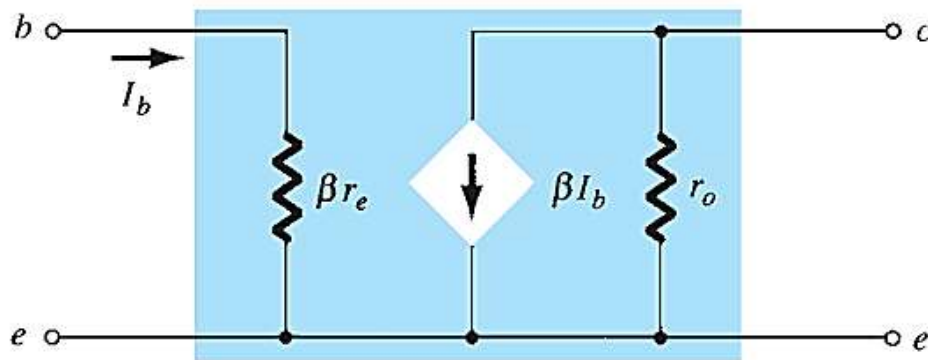
$$r_e = \frac{26 \text{ mV}}{I_E}$$



The r_e Transistor Model (Common Emitter Configuration)

$$\text{slope} = \frac{\Delta I_C}{\Delta V_{CE}} = \frac{1}{r_0}$$

$$r_0 = \frac{\Delta V_{CE}}{\Delta I_C}$$



COMMON-BASE CONFIGURATION

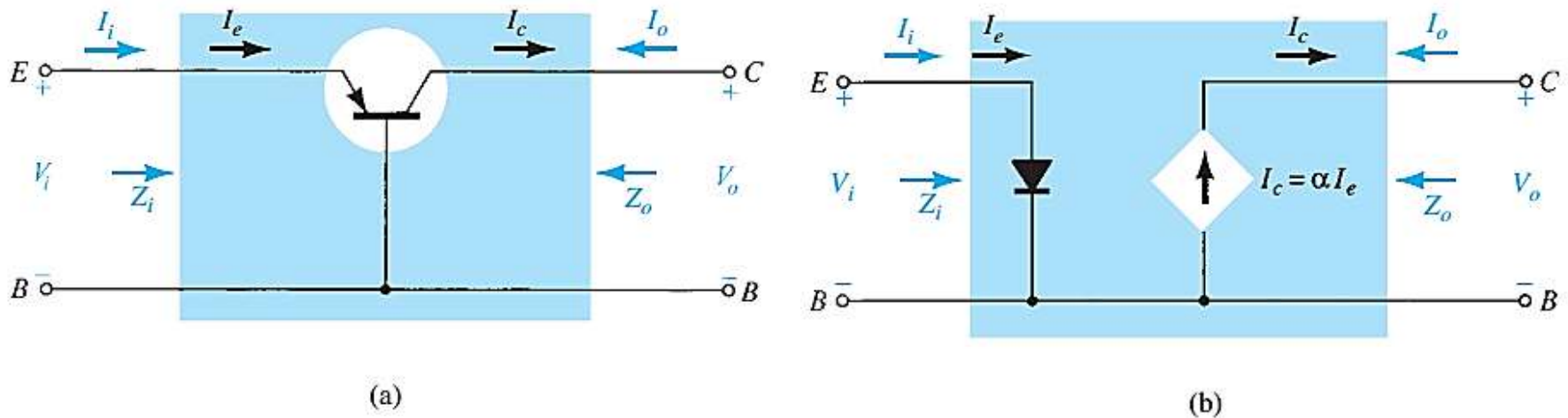


FIG. 5.17

(a) Common-base BJT transistor; (b) equivalent circuit for configuration of (a).

COMMON-BASE CONFIGURATION

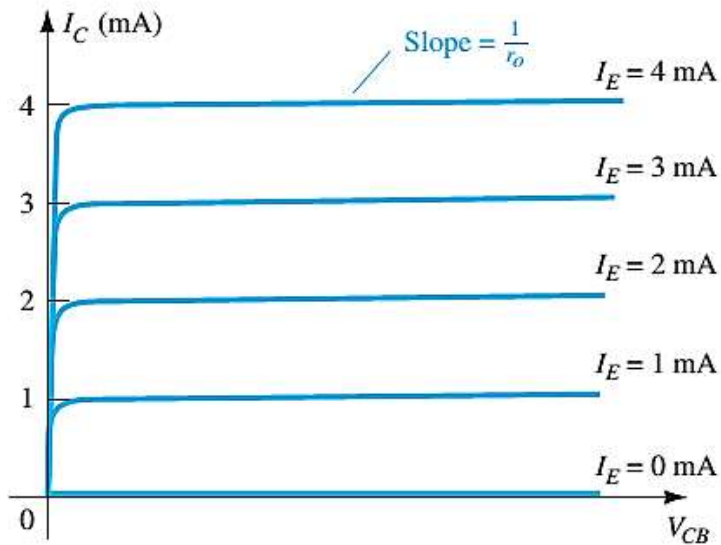
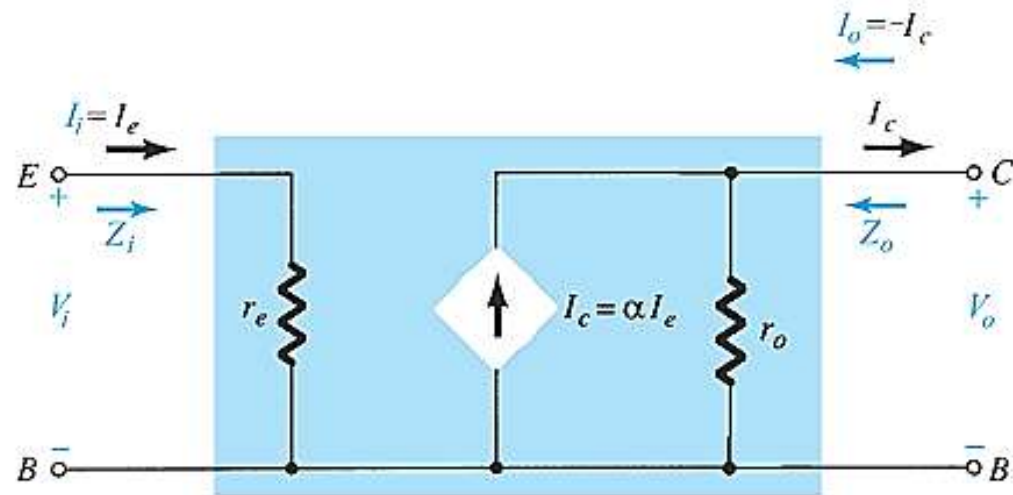


FIG. 5.19
Defining Z_o .



The output resistance r_o is quite high. typically extend into the $M\Omega$ range.

Common Base r_e
equivalent circuit

COMMON EMITTER FIXED BIAS CONFIGURATION

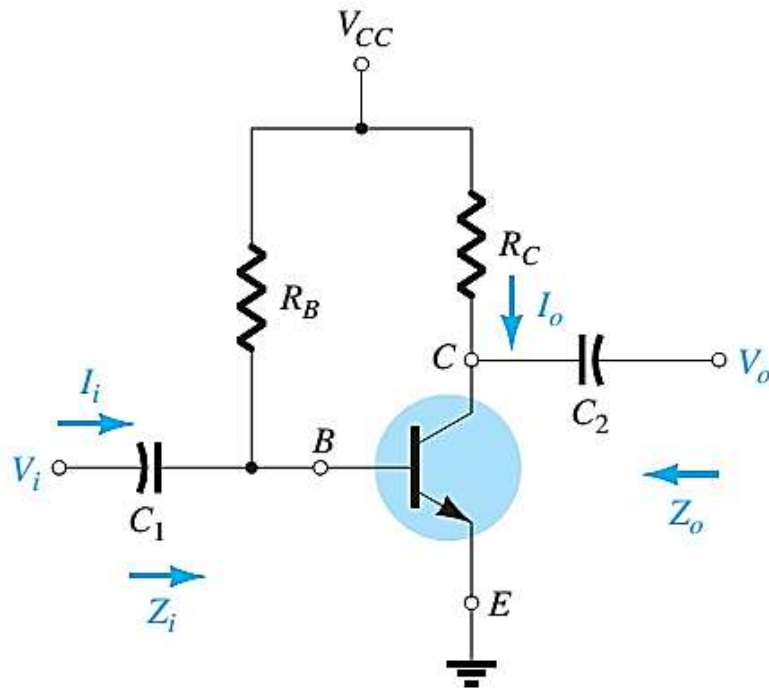


FIG. 5.20

Common-emitter fixed-bias configuration.

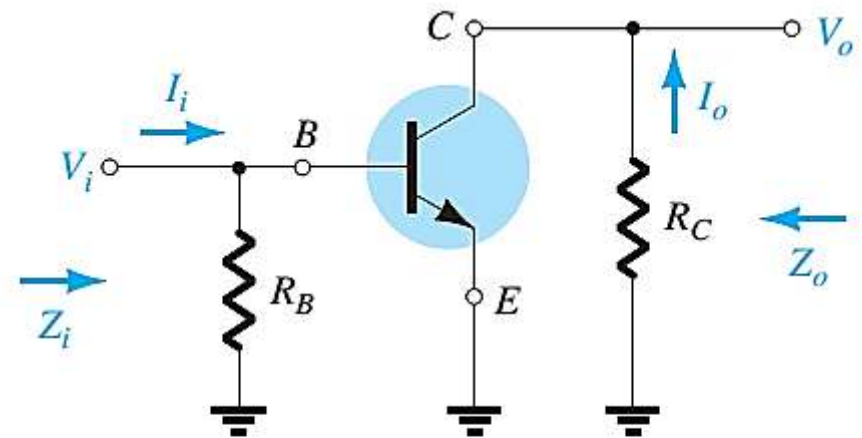


FIG. 5.21

Network of Fig. 5.20 following the removal of the effects of V_{CC} , C_1 , and C_2 .

COMMON EMITTER FIXED BIAS CONFIGURATION

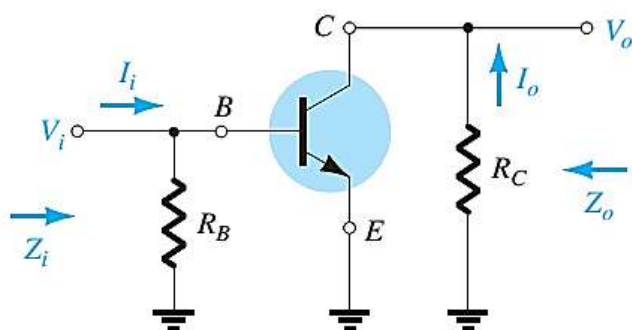


FIG. 5.21

Network of Fig. 5.20 following the removal of the effects of V_{CC} , C_1 , and C_2 .

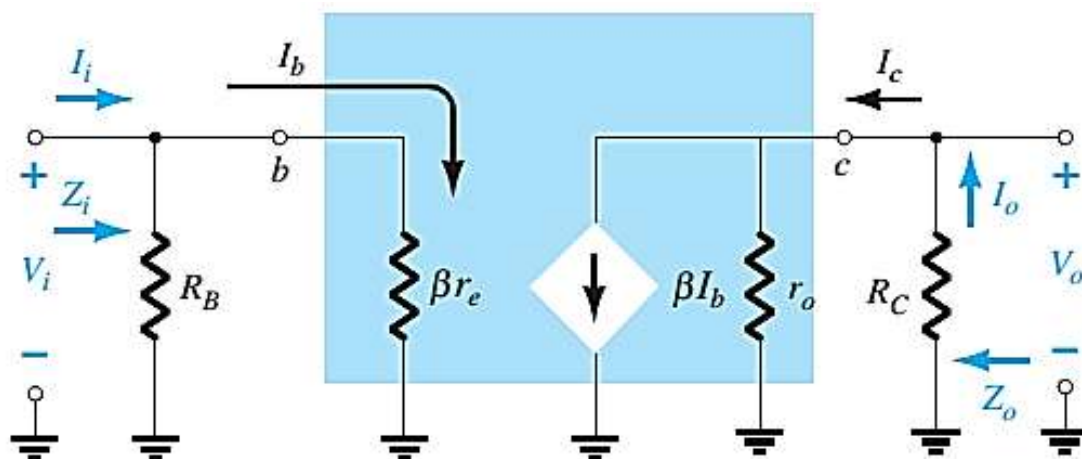


FIG. 5.22

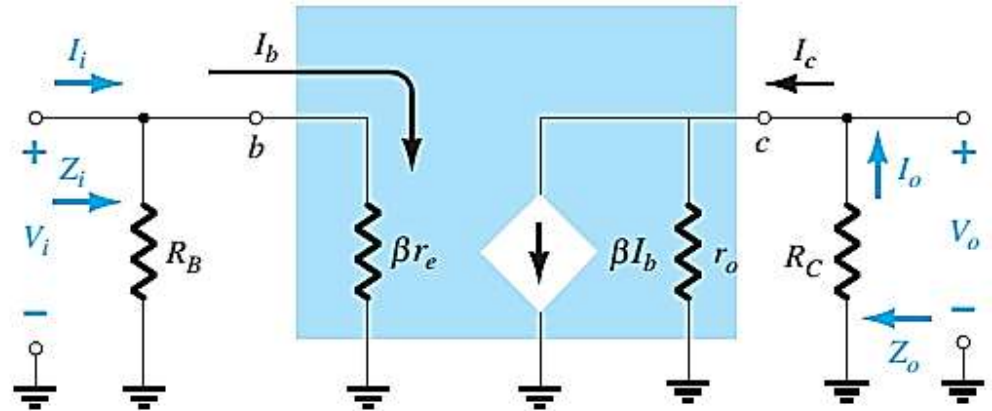
Substituting the r_e model into the network of Fig. 5.21.

COMMON EMITTER FIXED BIAS CONFIGURATION

INPUT IMPEDANCE, Z_i

$$Z_i = R_B \parallel \beta r_e$$

$$Z_i \cong \beta r_e \mid R_B \geq 10\beta r_e$$



OUTPUT IMPEDANCE, Z_o

$$Z_o = R_C \parallel r_o$$

$$Z_o \cong R_C \mid r_o \geq 10R_C$$

VOLTAGE GAIN, A_v

$$V_o = -\beta I_b (R_C \parallel r_o) = -\beta \left(\frac{V_i}{\beta r_e} \right) (R_C \parallel r_o); I_b = \frac{V_i}{\beta r_e}$$

$$A_v = \frac{V_o}{V_i} = -\frac{(R_C \parallel r_o)}{r_e}, A_v = -\frac{R_C}{r_e} \mid r_o \geq 10R_C$$

COMMON EMITTER FIXED BIAS PHASE RELATIONSHIP

$$A_v = \frac{V_o}{V_i} = - \frac{(R_C \parallel r_o)}{r_e}$$

Demonstrating the 180° phase shift between input and output waveforms.

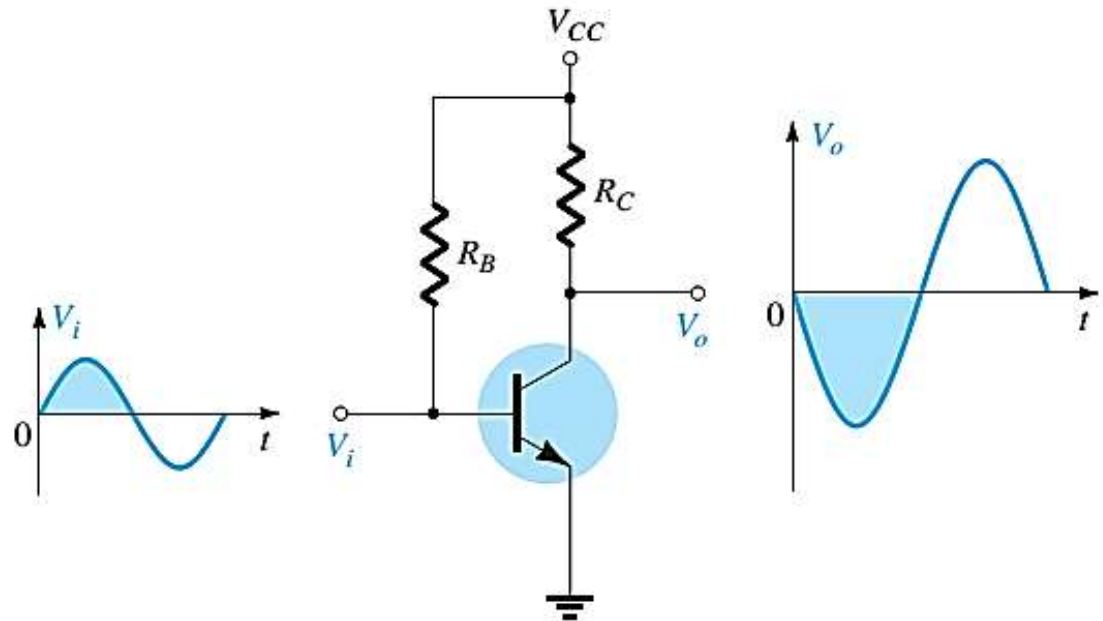


FIG. 5.24

Demonstrating the 180° phase shift between input and output waveforms.

EXAMPLE

- **EXAMPLE 5.1:** For the network of Fig. 5.25 :
- Determine r_e , Z_i (with $r_o = \infty$), Z_o (with $r_o = \infty$), A_v (with $r_o = \infty$) and Repeat with $r_o = 50 \text{ k}\Omega$.

a. DC analysis:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega} = 24.04 \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (101)(24.04 \mu\text{A}) = 2.428 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.428 \text{ mA}} = 10.71 \Omega$$

b. $\beta r_e = (100)(10.71 \Omega) = 1.071 \text{ k}\Omega$

$$Z_i = R_B \parallel \beta r_e = 470 \text{ k}\Omega \parallel 1.071 \text{ k}\Omega = 1.07 \text{ k}\Omega$$

c. $Z_o = R_C = 3 \text{ k}\Omega$

d. $A_v = -\frac{R_C}{r_e} = -\frac{3 \text{ k}\Omega}{10.71 \Omega} = -280.11$

e. $Z_o = r_o \parallel R_C = 50 \text{ k}\Omega \parallel 3 \text{ k}\Omega = 2.83 \text{ k}\Omega$ vs. $3 \text{ k}\Omega$

$$A_v = -\frac{r_o \parallel R_C}{r_e} = \frac{2.83 \text{ k}\Omega}{10.71 \Omega} = -264.24 \text{ vs. } -280.11$$

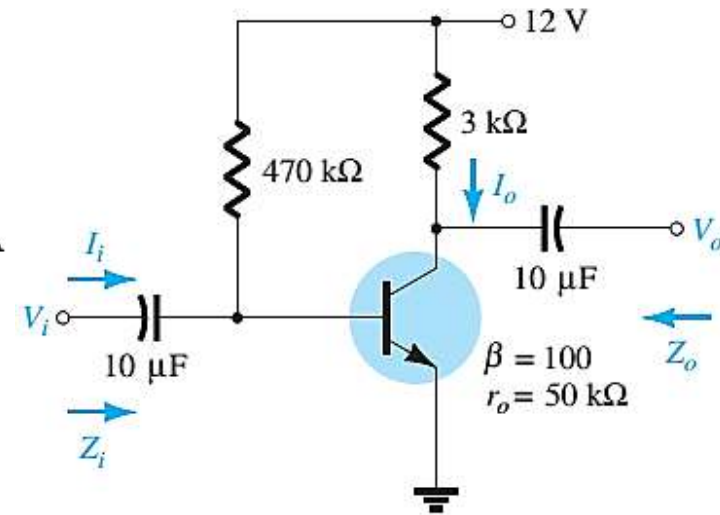


FIG. 5.25

Example 5.1.

End of Lecture